

High Frequency Groundwater Level Fluctuations, Underground Erosion, & Potential Sinkhole Occurrences Across Long Island, NY

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Abstract:

Sinkhole formations pose significant hazards to infrastructure and safety, with Long Island, NY experiencing a recent surge in such occurrences. This study investigates the potential correlation between high-frequency groundwater level (GWL) fluctuations and accelerated underground erosion, leading to sinkhole formation. Utilizing statistical analyses of GWL data from USGS monitoring stations, we identify locations exhibiting various GWL fluctuations across Long Island. While the results do not indicate a strong correlation between high-frequency GWL fluctuations and sinkhole occurrences on an individual basis, regions characterized by predominantly high-frequency GWL oscillations tend to have a higher incidence of sinkhole formations. This study provides insights into the factors contributing to sinkhole occurrences on Long Island and underscores the importance of further research in understanding and predicting such phenomena related to oncoming warmer temperatures and underground water mobility upon intensive climate change.

Introduction:

- LI groundwater is located in aquifers; Upper Glacial aquifer is closest to surface
- Groundwater level (GWL) fluctuations are influenced by natural forces and human activities
- Underground erosion is a result of groundwater flow through porous sediments
- Sinkholes form when the surface above underground erosion is destabilized and collapses inwards, this creates environmental hazards
- Previous studies on sinkhole formation on LI have not been widely investigated

HYPOTHESIS: Groundwater stations exhibiting high frequencies of GWL oscillations is an identifier of accelerated underground erosion, and thus an area prone to occurrences of sinkhole formations across Long Island, New York.



Figure 2: A Google Earth map showing the locations of sinkhole formations from 2013-2023 on Long Island, NY. Location data was sourced from local news reports.

Methodology:

- GWL data was downloaded from 58 USGS stations across LI
- Map of sinkhole formations on LI was created in Google Earth Pro
- GWL oscillations were analyzed in R using the following statistical tools:
 - Moving average
 - Kolmogorov-Zurbenko (kz) filter from the kza package (Close et al. 2020)
 - Periodogram time-series analysis (TSA)
- Map of stations with high-frequency GWL fluctuations were mapped using the following R packages:
 - Sf (Pebesma & Bivand, 2023)
 - ggspatial (Dunnington, 2023)
 - ggplot2 (Wickham, 2016)
 - Celestial (Robotham, 2018)

Conclusions:

- Identified a distinct relationship between shallow depth to water table and sinkhole formation occurrences
- High-frequency GWL oscillations occur at shallower depths to the water table on LI. These areas should be marked for further study due to any climatic conditions influencing GWL fluctuations to prevent flooding or sinkhole hazards.
- The influence of adjacent fault regions near LI on underground erosion and sinkhole formation requires further studies in order to identify relationships.
- Sinkhole occurrences are moderately related to high-frequency GWL oscillations at the USGS stations on LI.
- We can predict the relative location of future sinkhole formations by looking at the USGS stations showing high frequency GWL oscillations.
- Future studies would yield more coherent results for our hypothesis.

Results:

- Sinkholes mostly occur on the southwestern parts of LI with some exception
- We observe a moderate spatial correlation between sinkhole occurrence and USGS stations showing high-frequency GWL oscillations

Table 1: Sinkhole formations on Long Island, NY 2013-2023 and their respective locations and depths.

Sinkhole	Longitude	Latitude	Year	Depth (ft)
1) Huntington Station	-73.411268°	40.853503°	2023	6
2) Bay Ridge	-74.018219°	40.633155°	2024	20
3) Islip	-73.235022°	40.740219°	2014	12
4) Elmont	-73.712909°	40.700940°	2023	9
5) West Hempstead	-73.650142°	40.704847°	2023	6
6) Baldwin	-73.610711°	40.671756°	2023	n/a
7) Oceanside	-73.633872°	40.645077°	2023	20
8) Lido Beach	-73.629828°	40.587992°	2023	20

Discussion:

- Figure 1, 2, & 3: Sinkhole formations are located at shallow depths to the water table, < 30 ft below surface.
- Figure 2: observed a slight linear pattern of sinkhole occurrences in Nassau County. Lack of previously mapped geological structures suggest sinkhole formation may be more anthropogenically influenced.
- Figure 3: Sinkhole formations on LI are moderately correlated with USGS station sites showing high-frequency GWL oscillations. Low periodicity indicates higher frequency of GWL fluctuations and higher frequency of oscillations.

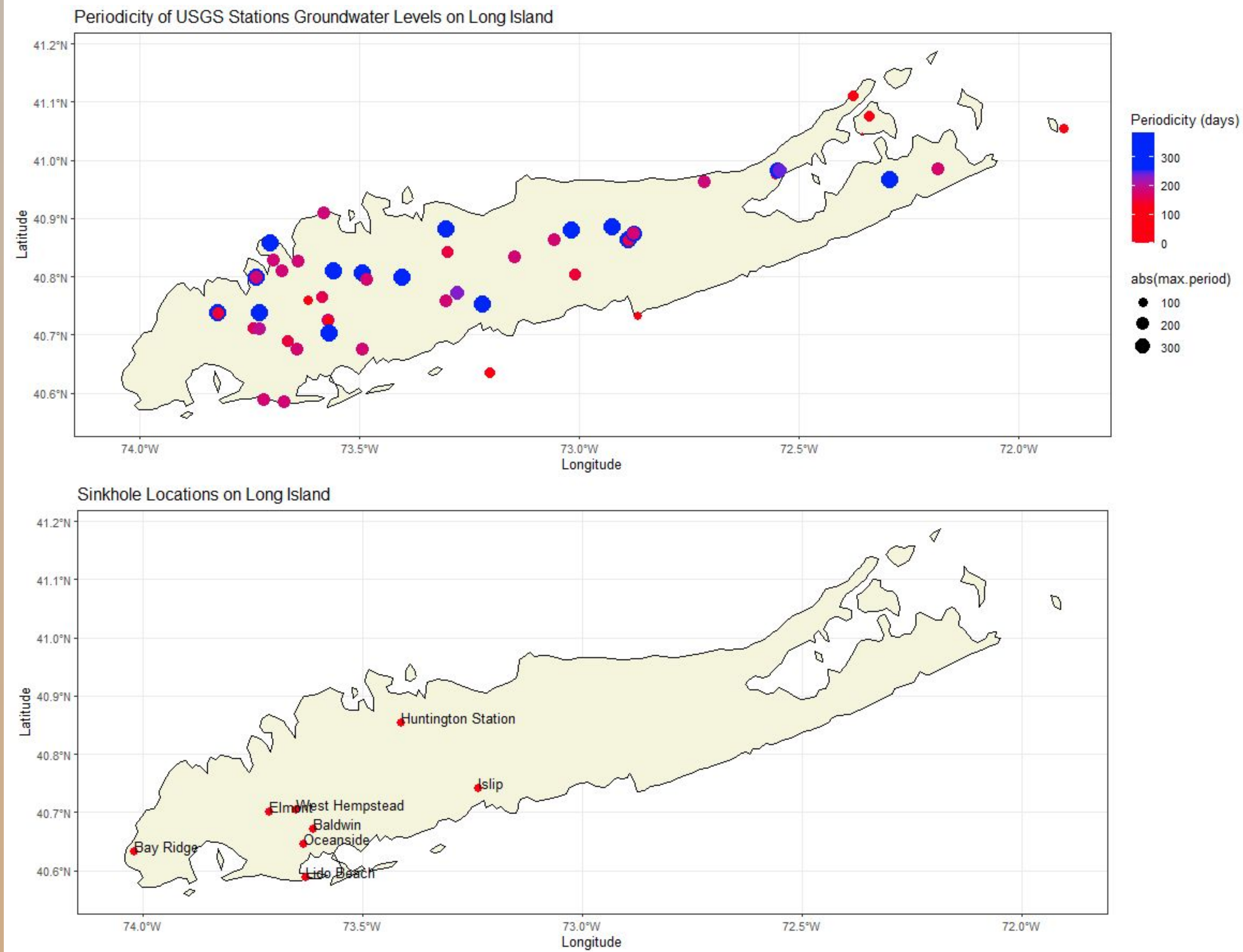


Figure 3: (Top) An Rstudio constructed GIS map showing the periodicity in days of USGS stations GWL across Long Island, NY. Higher frequency of GWL oscillations indicate lower periodicity (red dots). (Bottom) An Rstudio map of the sinkhole locations from Table 1 across Long Island, NY.

Credit Authorship & Contribution Statement:

Hope, J.H: writing – Introduction, Methods, Results, Discussion & Conclusion. Data analysis and R-coding practicing, reference citations, cartography, editing, formatting. Marsellos, A.E: writing – Abstract. Supervision, editing, GIS mapping, R-coding guidance. Tsakiri, K.G: R-code/providing additional insights to KZ filter for data processing.

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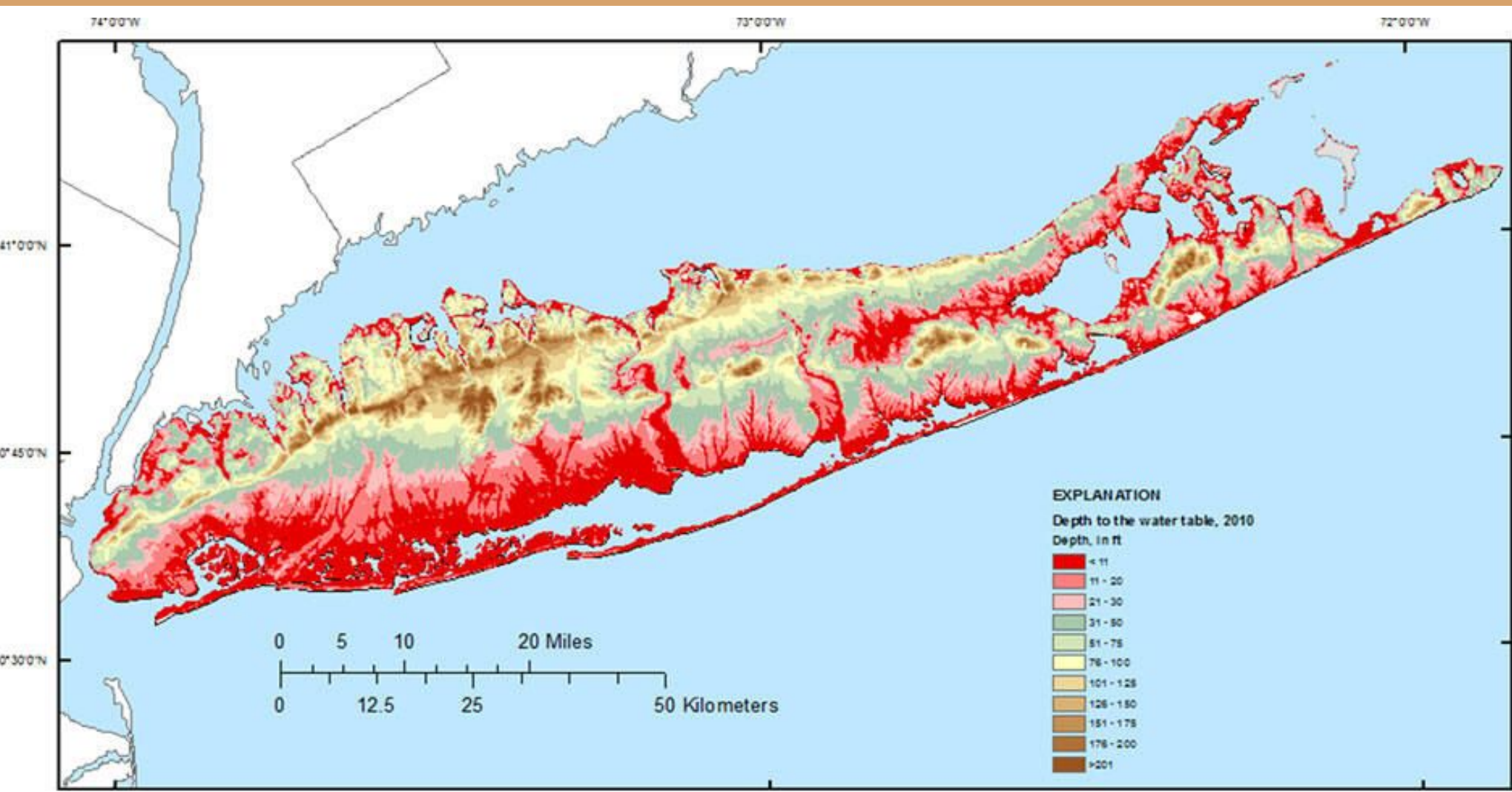


Figure 1: USGS map of the depth to the water table in feet on Long Island, NY 2010 (Como et al. 2013). Red indicates < 11 ft depth.